

Department of Electrical and Computer Engineering
University of Wisconsin – Madison
ECE 552 Introduction to Computer Architecture
In-Class Exercise (09/17)

1. Write MIPS assembly code that sets the contents of register \$t0 to the value 262,158 (in decimal). Note that the number of bits needed to encode this number is **greater than 16**.

Answer:

lui \$t0, 4
ori \$t0, \$t0, 14

2. You propose an improvement to your processor that accelerates all arithmetic instructions by 3x. If three-fourths of your program's runtime is spent on arithmetic instructions, compute the expected speedup on the total program runtime using **Amdahl's Law**.

Answer:

$1 / (1 - f + f/3)$, where f is $3/4$, = 2x speedup.

3. Consider a program running on your computer. You apply a technique for reduced precision that accelerates non-memory instructions in your program by 4x. Your measurements then show that this speeds up the entire program execution by 3x. Now in addition to this reduced-precision technique, would it be better for performance to either (i) introduce a new technique on top of reduced-precision to speed up all **memory** instructions by 8x, or (ii) improve the reduced-precision technique such that it accelerates **non-memory** instructions by 8x instead of 4x?

Answer:

$\frac{1}{1-f+\frac{f}{3}} = 3$, where f is the fraction of instructions that are non-memory.

Thus $f = \frac{8}{9}$ is non-memory.

(i) $\frac{1}{\frac{1/9 + 8/9}{8 + 4}} = 4.24x$

(ii) $\frac{1}{\frac{1}{9} + \frac{8}{8}} = 4.5x$

Thus (ii) would be better

4. Consider a processor architecture with an average CPI of 1.75. You propose a change to the architecture that can completely eliminate **half** of all memory read instructions in the program but leads to an average **CPI of 2.2** instead. Given the typical instruction breakdown that you expect in your program (table below), is this change a worthy trade-off for performance?

% of Program	Instruction Type
4%	addition/subtraction
10%	multiplication
38%	memory read
24%	memory write
8%	logic
16%	branch

Answer:

Program runtime, via Iron Law = (insns) × (CPI) × (clock period)

Since $(\text{insns} \times (1 - \frac{0.38}{2}) \times (\text{CPI} = 2.2) > (\text{insns}) \times (\text{CPI} = 1.75)$, this would not be a worthy trade-off.